Al–Cu liquidus
where the last solid disappears

/machine Al-Cu alasımı kaç derecede tamamen erir?

LIQUIDUS · ALLOY

927.78 K

COMPOSITION
Cu 2.0 wt%

LEDGER
20261026T233221-435ddb5d

GATE
PASS · 0.342 K bracket

VERDICT
GREEN · CALPHAD equilibrium

21 BAKED POINTS · THE CURVE IS THE DATABASE, THE DOTS ARE LEDGER ENTRIES



x = Cu wt% · y = liquidus K · the highlighted dot is the number above · no point on this line is computed in the page

Cu wt%	liquidus K	± K	ledger
0.0	933.47	0.043	20261026T233204-f40e097a
0.5	931.88	0.342	20261026T233208-e570ada1
1.0	930.52	0.342	20261026T233213-ecdcb4c8
1.5	929.15	0.342	20261026T233217-6b962052
2.0	927.78	0.342	20261026T233221-435ddb5d
2.5	926.42	0.342	20261026T233225-97f3805d
3.0	925.05	0.342	20261026T233230-d31dcae2
3.5	923.00	0.342	20261026T233234-9387a0c5
4.0	921.63	0.342	20261026T233238-b02945d1
4.5	920.26	0.342	20261026T233242-ce0052f1
5.0	918.90	0.342	20261026T233246-f7643683

Cu wt%	liquidus K	± K	ledger
5.5	917.53	0.342	20261026T233251-a85ac4b0
6.0	916.16	0.342	20261026T233254-236b2af9
6.5	914.11	0.342	20261026T233258-4dcb762e
7.0	912.74	0.342	20261026T233303-c770ed5d
7.5	911.38	0.342	20261026T233307-13cffd28
8.0	910.01	0.342	20261026T233311-40c268a7
8.5	908.64	0.342	20261026T233315-3822b2c5
9.0	906.59	0.342	20261026T233319-ff039e6d
9.5	905.22	0.342	20261026T233323-c4b77968
10.0	903.86	0.342	20261026T233327-a4720ad1

HOW IT IS COMPUTED

- x(Cu) from wt%: n_{Cu} = w/63.546, n_{Al} = (100−w)/26.9815, x = $n_{Cu}/(n_{Cu}+n_{Al})$
- liquidus: bisect T where any solid phase has NP > 1e−6 (9 steps), bracket reported
- x_{Cu} = 0: no bracket exists, pure-element path (FCC_A1 → LIQUID)

WHAT THIS DOES NOT SAY

- liquidus of 2.0 wt% Cu = mole fraction x(Cu) = 0.00859; Al-Cu database phases only
- ±0.342 K is the bisection bracket (numerical resolution), not database or model uncertainty

BREAK IT · OUT OF DOMAIN

no data · gate **RED** · Au — no ELEMENT entry

Bu madde/veri seti elimizde yok; en yakin kapsam: alloy.phase

ledger 20261026T225932-91f72739
the sheet shows the refusal, not a number

Reading the sheet

The number on top is not a fit. It is the temperature at which pycalphad finds the last solid fraction disappearing in the MatCalc Al-Cu database, bisected at that composition and stored as one ledger entry. Each row below is the same procedure at another composition; the curve is drawn through those rows, not through a formula.

The page behind this sheet snaps to the same 21 points. Between them the page model interpolates linearly inside a declared ±0.5 K, and **lab_audit.py** checks every midpoint against the engine: worst case measured **0.3418 K**.

Why this is not a simulation

The database is an assessment of published Al–Cu thermodynamics (mc_al_v2.032). The ±0.342 K is the bisection bracket — how tightly the solver closed in — not the uncertainty of the database. A real measurement of this liquidus would carry its own error, usually several kelvin, and that error is not claimed here.

The rig: 52 named parts in five groups (base, structure, mechanism, controls, subject). The cutaway opens the top-front quarter so the quartz crucible, the six-turn coil and the type-K sheath are visible; nothing in the scene is a decal.

Provenance · pycalphad equilibrium (Al-Cu liquidus) · mc_al_v2.032.pycalphad.tdb
baked 2026-10-27 02:33:27 · repo head 2e21943 · model al_cu_liquidus_r3.json
regenerate: **python scripts/gen_model.py** · audit: **python scripts/lab_audit.py --compute**

lab kit: AirsupHQ/machine-skill (MIT)
page model ±0.5 K between anchors
anchor = ledger · midpoint tested